

Hao Nguyen The

Backend Developer

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PROFILE

Mid-level Backend Developer with 2.5+ years building scalable systems across e-commerce, fintech, and government sectors. Strong in NestJS, PostgreSQL, Redis, and microservices architecture. Experienced with async pipelines, eKYC workflows, external identity-provider integrations, and automated data crawling. Targeting Mid-level roles in Vietnamese and international markets.

EXPERIENCE

Backend Developer — Trident Digital Tech

Jul 2025 – Present · Ho Chi Minh City

RDC — National Citizen Management Platform (DR Congo, 100M+ population)

- National-scale system needed a resilient backend; **architected NestJS microservices** separating core APIs, async eKYC workers, and cron jobs — designed for 100M+ citizens.
- Large citizen datasets caused DB bottlenecks; applied **Redis + Bloom Filter** caching and **BullMQ** async queues, cutting DB load and decoupling OCR/face recognition from the API cycle.
- Identity verification required a secure, auditable flow; built **eKYC workflow** with Tencent Cloud OCR & Face Recognition, JWT auth, RBAC, and AES encryption.
- Citizens had to be reconciled against national registries; built **verification-source linking** against external providers (ONIP, CENI) with auto/manual linking, duplicate-prevention, and a two-step manual enrollment flow.
- Fraud & abuse risks on lookups; added **rate-limiting** on national-ID enquiry and per-API permission configuration (RBAC).
- Records needed flexible search and full auditability; configured **Elasticsearch** full-text search, designed **PostgreSQL** schemas with migrations, and streamed user-event logs to **ClickHouse** for the analytics/audit trail.

Trifood — Food Delivery & Surplus Marketplace

- APIs needed to be reliable and well-tested; built the **Surplus Bag** module and merchant order/product/payment endpoints (NestJS) with cart & order validation, achieving **70%+ test coverage** via Jest.
- Surplus inventory and variant pricing had to stay consistent; implemented auto-sold-out status, variant price sync on markup changes, and a **BullMQ + Redis** processing pipeline.
- Product search was slow on large catalogs; indexed data in **Elasticsearch** to power fast full-text queries across the product catalog.

Software Developer — Tego Global

Dec 2023 – Jun 2025 · Ho Chi Minh City

Pfs — Automated Insurance Data Crawling

- Manual collection took ~5 min/record, unscalable for thousands of entries; built **Scrapy + Selenium** crawlers (17 spiders) reducing time to **<30s/record (10x+ faster)**; API sources completed in <3s.
- Anti-bot protection blocked scraping; bypassed **Cloudflare** with cloudscraper/cfscape and rotating user-agents, scaling to **tens of thousands of records/day** across motor, home, travel & pet insurance (AAMI, Allianz, QBE, CBA, NIB, Covermore...).
- Crawlers needed orchestration and delivery; built a **NestJS backend** driving the Python crawler over **gRPC**, with real-time scrape progress via **Server-Sent Events (SSE)**, TypeORM migrations, and JWT + OTP + login-attempt auth with RBAC.
- Needed production infra; used **AWS S3, Cognito, RDS**, managed ORM via TypeORM & SQLAlchemy (Alembic migrations), and Redis caching.

Perkify — CRM & Loyalty System (Vivaleisure)

- CRM lacked a rewards engine; built **loyalty point & redemption** features on **NestJS + MongoDB**, optimizing aggregation queries for faster reporting.
- Multiple roles and channels; developed REST APIs for admin, staff & customers with JWT + Passport (**Facebook social login**), **WebSocket** real-time updates, rate-limiting (Throttler), and cron-driven **nodemailer** email campaigns.
- Deployment was manual and error-prone; provisioned cloud infrastructure as code with **AWS CDK** (EC2, ECR, ELB, IAM, Lambda, Route 53).

Blog & Autosend News — Internal Platform

- ~2 hrs/day lost to manual content curation; built **automated Slack integration** via cron jobs delivering tech news & crypto updates, **saving ~2 hrs/day**.
- Platform lacked real-time features; implemented WebSocket notifications, tree comments, content recommendations (Embeddings), Redis caching — deployed on NestJS + Prisma + PostgreSQL.

SKILLS

Backend NestJS, Node.js, Next.js, gRPC, WebSocket/SSE
Languages TypeScript, JavaScript, Python
Databases PostgreSQL, MongoDB, Redis, Elasticsearch, ClickHouse

Cloud AWS S3, EC2, RDS, Cognito, CDK (IaC)
Infra Docker, GitHub Actions
Messaging BullMQ, Redis Queues
Testing Jest (Unit & E2E)

EDUCATION & LANGUAGES

B.Eng. Software Engineering

HCMUTE, Ho Chi Minh City · Aug 2021 – June 2026

Vietnamese — Native

English — TOEIC 740